



Cyberbullying in adolescents: Modalities and aggressors' profile

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ABSTRACT

In this study, a questionnaire (Cyberbullying Questionnaire, CBQ) was developed to assess the prevalence of numerous modalities of cyberbullying (CB) in adolescents. The association of CB with the use of other forms of violence, exposure to violence, acceptance and rejection by peers was also examined. In the study, participants were 1431 adolescents, aged between 12 and 17 years (726 girls and 682 boys). The adolescents responded to the CBQ, measures of reactive and proactive aggression, exposure to violence, justification of the use of violence, and perceived social support of peers. Sociometric measures were also used to assess the use of direct and relational aggression and the degree of acceptance and rejection by peers. The results revealed excellent psychometric properties for the CBQ. Of the adolescents, 44.1% responded affirmatively to at least one act of CB. Boys used CB to greater extent than girls. Lastly, CB was significantly associated with the use of proactive aggression, justification of violence, exposure to violence, and less perceived social support of friends.

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1. Introduction

Violent behavior among adolescents and young people is a severe problem in many countries. In recent years, new forms of aggression based on information and communication technology (computers, cell phones, etc.) have been added to the traditional forms of violence. In this context, cyberbullying (CB) has been defined as an aggressive and deliberate behavior that is frequently repeated over time, carried out by a group or an individual using electronics and aimed at a victim who cannot defend him- or her-self easily (Smith, 2006). Patchin and Hinduja (2006) describe it as deliberate and repeated harm performed with some kind of electronic text. These violent behaviors can be carried out by means of a cell phone, electronic mail, Internet chats, and online spaces such as MySpace, Facebook, and personal blogs.

Although in many cases, CB implies acts of traditional aggression (for example, insulting, spreading rumors, or threatening), which are communicated electronically instead of face-to-face, CB can also include unique behaviors with no analogue in traditional bullying. For example, the phenomenon known as *bombing* occurs when the aggressor uses an automated program to collapse the victim's e-mail with thousands of simultaneous messages, causing failure and blocking of the victim's e-mail account (Burgess-Proctor, Patchin, & Hinduja, 2008).

As this phenomenon is new, there is as yet little agreement about the diverse categories of this form of violence, so that in the studies carried out, different classifications can be found (e.g., Burgess-Proctor et al., 2008; Smith, Mahdavi, Carvalho, & Tippett, 2006; Willard, 2006, 2007). For example, according to Willard (2006, 2007), some of the modalities that CB can adopt are (1) on-line fights, known as *flaming*, which imply the use of electronic messages with hostile and vulgar language; (2) slandering, a modality that implies online disparagement, for example, sending cruel images or rumors about others to spoil their reputations or social relationships; (3) impersonation (hacking) by infiltration into someone's account in order to send messages that make the victim lose face, cause trouble for or endanger the victim, or harm the victim's reputation and friendships; (4) defamation by spreading secrets or embarrassing information about someone; (5) deliberate exclusion of someone from an online group; (6) cyber harassment or the repeated sending of messages that include threats of injury or that are very intimidating.

The phenomenon known as *happy slapping* consists of recording with cell phone cameras images in which a person, who is often in a minority situation, is attacked. The image or video is later shared with friends, posted online, or distributed electronically. This phenomenon has recently been the object of attention by the mass media in many European countries.

Diverse studies warn about the high occurrence of CB. Table 1 shows a sample of some representative studies. For example, in one of the first studies of CB, Ybarra and Mitchell (2004) surveyed 1501 children and adolescents between ages 10 and 17 years by phone and found that 12% had participated in cyberbullying. Li

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Table 1
Representative studies of cyberbullying.

Authors	Measurement of CB	Sample	Prevalence rate
Aricak et al. (2008)	"Questionnaire of Cyberbullying" (QoCB)	269 students, aged between 12 and 19 years from Istanbul	35.7%
Beran and Li (2005)	Questionnaire of 15 items, based on a definition of harassment. They use open questions to specify types. Closed for frequency, being victim/aggressor, emotional and behavioral responses	432 students from 7th to 9th grade. Canada	22% once or twice; 4% several times. No gender differences
Dehue et al. (2008)	They created a questionnaire for students and one for parents. The questionnaire for students, among other things, asks about the prevalence and the methods used to carry out and/or be a victim of CB (SMS, e-mail, gossiping, ignoring, hacking, name-calling), the anonymity of the aggressor and the sex of the victim. The questionnaire is based on the based on "Olweus Bully/victim Questionnaire" and on the "Amsterdam Bullying Questionnaire for Children"	1211 participants from the last grade of primary education and the first of secondary education. Mean age: 12.7 Holland	16% significantly more boys (18.6%) in comparison with girls (13.4%)
Li (2006)	An anonymous survey that, among other aspects, asks whether the respondent had cyber-bullied others. If so, they should indicate the means (e-mail, chat room, cell phone, other) and the frequency	264 students from 7th to 9th grade. Canada	17% (22.3% boys and 11.6% girls)
Li (2007a)	The same questionnaire as in Li (2006)	177 Canadian students from 7th grade. Canada	14.5%
Li (2007b)	The same questionnaire as in Li (2006)	461 students from 7th grade. Canada and China	17.8% (21.9% boys and 13.4% girls)
Li (2008)	The same questionnaire as in Li (2006)	354 students between 11 and 15 years. Canada and China	Canada: 15% China: 7%
Ortega et al. (2008)	Cyberbullying Questionnaire (Ortega, Calmaestra, & Mora-Merchán, 2007). Includes questions about CB through cell phones and the Internet	830 Spanish students between 12 and 18 years	1.7% severe cyberbullying, 5.7% occasionally or moderate
Hinduja and Patchin (2008)	Online. First, a description of CB. Includes two questions: Have you ever performed CB against others? and Have you ever threatened to physically harm someone or have you scared others this way online?	680 boys/698 girls 10–17 years	No gender differences 18% boys and 15.6% girls
Patchin and Hinduja (2006)	Online. First, a description of CB. Includes two questions: Have you ever performed CB? and, Have you ever threatened to physically harm or scared others this way online?	384 people less than 18 years of age	10.7%
Smith et al. (2008)	Questionnaire based on Olweus' Bully/Victim Questionnaire (Solberg & Olweus, 2003). The questionnaire includes a definition of bullying followed by a statement about cyberbullying as including the seven media: through text messaging, pictures/photos or video clips, phone calls, e-mail, chat rooms, instant messaging, and websites. Next, it asks about the frequency of CB, for each of the seven media. Open-ended questions allowed pupils to give more detailed answers on examples of cyberbullying, reasons for perceived impact, and suggestions for stopping it. The time-frame was the "past couple of months"	Two surveys with pupils aged 11–16 years: (1) 92 pupils, supplemented by focus groups; (2) 533 pupils	6.6% often (2 or 3 times a month, once a week, or several times a week) and 15.6% once or twice No gender differences
Ybarra and Mitchell (2004)	"Youth Internet Safety Survey". By telephone. Included two questions for aggressors: (1) Have you ever made disagreeable or vulgar remarks about others on the Internet? (2) Have you used the Internet to harass or shame someone with whom you were angry?	1501 young people between 10 and 17 years USA	12%
Williams and Guerra (2007)	They use one item: I told lies about some students though e-mail or instant messaging	3339 youths in 5th, 8th, and 11th grades	9.4%

(2006), in a sample of 264 high school students, found that 22% of the boys and 12% of the girls admitted having cyberbullied others. In a cross-cultural study, the same authoress found percentages of cyberbullies ranging between 15% and 7%, respectively, for Canada and China (Li, 2008). In a study carried out with Turkish students, 35.7% admitted performing CB (Aricak et al., 2008). In Spain, Ortega, Calmaestra, and Mora-Merchán (2008) found in adolescents that 5.7% admitted having performed CB occasionally, and 1.7% had carried out severe forms of CB. Although most of the research was carried out in schools, in some cases it was online (Hinduja & Patchin, 2008).

A limitation of many of the studies is that they assessed the occurrence of CB generically, without specifying in detail the modalities employed. For example, in some cases, they focused on providing a definition of CB and asking the participants whether they had carried out CB, and if so, to describe aspects such as the means employed (chat room, e-mail, cell phone). This perspective is valuable to address a new phenomenon about which relatively few studies have been carried out. However, as we discover more about the importance of the phenomenon, it becomes appropriate to develop more specific measures that include a broad array of CB modalities. Therefore, the first goal of this study consisted of developing a questionnaire to assess the performance of many types of CB by adolescents.

On the other hand, CB reveals a series of differences with the traditional types of maltreatment and bullying among schoolmates. Elements such as the perception of online anonymity and the safety of hiding behind a computer screen contribute to freeing individuals from traditional constraints and social pressures, as well as from moral and ethical misgivings (Hinduja & Patchin, 2008; Li, 2007a, 2007b). Thus, adolescents who would not behave violently in a face-to-face situation can adopt different roles and perform this type of violence. Anonymity also implies the absence of consequences, because the aggressors frequently cannot be identified and, therefore, they avoid punishment. These characteristics make one wonder whether the psychological profile of the adolescents who carry out CB is similar to or different from the profile associated with traditional forms of violence. Therefore, the second goal of the study consisted of assessing the relation between CB, offline forms of violence, and associated risk factors.

Regarding the relation of CB to other violent behaviors in adolescents, CB should be associated with forms of proactive aggression and indirect aggression. Proactive aggression consists of deliberate and planned behavior with the intention of obtaining a reward and is differentiated from reactive aggression, which refers to a furious response to a perceived threat or provocation (Dodge, 1991). In fact, previous studies suggest that traditional bullying is more closely associated with proactive than with reactive aggressiveness (Roland & Idsøe, 2001; Schwartz et al., 1998; Unnever, 2005). On the other hand, indirect aggression, also called relational or social aggression, consists of harming someone by means of manipulating relationships (Björkqvist, 2001; Björkqvist, Lagerspetz, & Kaukiainen, 1992; Crick & Grotpeter, 1995). In this type of aggression, covert strategies are used in order to exclude and isolate rivals in the peer group. These actions include spreading rumors about others, threatening to end personal relationships, and to reveal private information (Crick, 1995; Galen & Underwood, 1997). In this sense, certain forms of CB have the same characteristics as traditional indirect bullying (Dehue, Bolman, & Völlink, 2008) and CB has even been defined as a computer-mediated form of indirect aggression (Piazza & Bering, 2009).

One of the risk factors that have been traditionally associated with the above-mentioned forms of aggressive behavior, and with bullying in particular, are normative beliefs about the justification of violence. Adolescents' experiences throughout their lives lead them to store in their memories certain knowledge structures that

affect their future behavior. Cognitive-social theories have generally called such knowledge structures schemas or scripts (Huesmann, 1988). In the case of aggressive behavior, many studies have revealed the presence of schemas related to the justification of the use of violence. For example, various studies have detected that children and adolescents who believe that it is appropriate to attack others when they deserve it are more apt to be aggressive (Bentley & Li, 1995; Bosworth, Espelage, & Simon, 1999; Calvete, 2008; Calvete & Cardeñoso, 2005; Huesmann & Guerra, 1997). In fact, a recent study of Williams and Guerra (2007) found an association between justification of violence and CB. However, in their study, the measurement of CB was a single question that referred to spreading lies about classmates by e-mail or instant messaging, and other forms of CB were not included.

In addition, a series of contextual variables have been linked to violent behavior for a long time. Firstly, the role of exposure to violence was pointed out from the social learning model by Bandura (1986). Children who observe more positive consequences and fewer negative ones for aggression acquire the belief that aggressive behavior leads to good consequences. In general, diverse studies support the fact that aggressive behavior increases with exposure to violence at home, at school, in the neighborhood, and in the mass media (Baldry, 2003; Coyne & Archer, 2005; Flannery, Wester, & Singer, 2004; Gorman-Smith, Henry, & Tolan, 2004; Harold & Conger, 1997; Huesmann, Moise-Titus, Posolski, & Eron, 2003; Schwartz & Proctor, 2000).

Lastly, among the contextual variables are experiences with peers, and rejection by peers is one of the most important factors (Laird, Jordan, Dodge, Pettit, & Bates, 2001). Numerous investigations have found a clear relation between aggressive behavior and rejection (see Dodge, Coie, & Lynam, 2006). However, the results have been mixed, depending on the type of aggressive behavior (Card & Little, 2006; Price & Dodge, 1989; Salmivalli, Kaukiainen, & Lagerspetz, 2000), and, in general, suggest that rejection is positively associated with reactive aggression and negatively with indirect aggression.

Summing up, the goals of this study were (1) to develop a questionnaire to assess a variety of CB behaviors. This goal in turn implied the assessment of the measurement model of the instrument. (2) To study the relation of CB with other indicators of aggressive behavior such as the justification of violence and the frequency of proactive, reactive, direct, and indirect aggressive behaviors. And (3) to study the association between CB and diverse contextual variables such as exposure to violence and acceptance and rejection by peers.

2. Method

2.1. Participants

The sample was made up of 1431 adolescents, between 12 and 17 years, high school students from 31 classrooms of 10 educational centers of Bizkaia (Spain). The measurements were taken between March and May of 2008. Of the sample, 726 were girls, 682 boys, and 23 did not indicate gender. The mean age was 14.09 years ($SD = 1.33$). The degree of representativeness of the sample was 3.82% and the sampling error 2.6%. A cluster-sampling method was used, and high schools were selected randomly. Of the adolescents, 94.8% were White and 5.2% of other ethnic minorities. Regarding academic course, 353 were first-graders of secondary education, 353 were in second grade, 369 in third grade, and 300 were in fourth grade. To determine the parents' socioeconomic level, we followed the recommendations of the Work Group of the Spanish Society of Epidemiology and the Spanish Society of Family and Community Medicine (2000), which are based on the last job

held by the parents. According to this criterion, the distribution was as follows: 14.4% low, 20.7% low-medium, 30.3% medium, 21.2% high-medium, and 13.3% high.

2.2. Measures

2.2.1. The Cyberbullying Questionnaire (CBQ)

This questionnaire was developed in this study. The CBQ consists of 16 items that describe 16 forms of CB, such as sending threatening or intimidating messages to someone, impersonating someone (hacking), recording aggressions by cell phone, excluding an online companion, etc., which were identified in previous studies (Smith et al., 2006; Willard, 2006, 2007). Some of the items imply very active participation (e.g., hanging intimidating material about a classmate on the Internet), whereas others assume collaboration with the previous action (e.g., resending the link to other people for them to see). The adolescent had to respond on a scale of 0 (*never*) 1 (*sometimes*), and 2 (*often*). In addition, various items include open questions to describe the behavior carried out (e.g., Item 3: *In the case of hanging humiliating images of a classmate on the Internet, describe what kind of images*). The Appendix includes the complete questionnaire.

2.2.2. The Irrational Beliefs Scale for Adolescents (IBSA; Cardeñoso & Calvete, 2004)

The Justification of Violence subscale of the IBSA consists of nine items that reflect the idea that aggression is appropriate in a variety of situations (e.g., “Sometimes you have to hit others because they deserve it”), and that aggression enhances self-esteem and helps to maintain status among peers (e.g., “Being good at fighting is something to be proud of” and “It is better to have a row than let them think I am a coward”). Each item is rated on a 4-point Likert-type scale ranging from 1 (*not at all true*) to 4 (*completely true*). Previous research has shown that this subscale is associated with aggressive and delinquent behavior (Calvete & Cardeñoso, 2005). In this study, the alpha coefficient was .84.

Multiple measures were employed to assess aggressive behavior. The *Reactive–Proactive Aggression Questionnaire* (Raine et al., 2006) consists of 11 items for reactive aggression (“How often have you become angry or mad when you don’t get your way?”) and 12 items for proactive aggression (“How often have you had fights with others to show who was on top?”). Each item is rated as 0 (*never*), 1 (*sometimes*) or 2 (*often*). The questionnaire has shown good reliability and validity in adolescent samples. Alpha coefficients in this study were .81 and .85 for reactive and proactive aggression, respectively. In addition, a peer nomination instrument, similar to that used in other studies (Crick & Grotpeter, 1995), was employed to assess direct and indirect aggressiveness. The instrument has 8 items, 4 for direct aggressiveness (e.g., classmates who hit or shove others) and 4 for relational/indirect aggressiveness (e.g., a classmate who tells his or her friends that they will no longer be friends unless they do what he or she wants). Previous studies have found satisfactory internal consistency of the scales (e.g., Crick, 1995; Crick & Grotpeter, 1995) and good test–retest validity, $r = .82$ for indirect aggressiveness and .90 for direct aggressiveness over a 4-week interval (Crick, 1996). Following the procedure employed in previous studies (Crick, 1995; Crick & Grotpeter, 1995), the adolescents were requested to name three classmates of either sex who had behaved as described. The number of nominations that the adolescents received from their classmates for each item was added and standardized in each class. Then, the standardized scores of each item for each subscale were added to obtain the total score of direct and indirect aggressiveness. The Cronbach alpha coefficients in this study were .90 for direct aggressiveness and .77 for indirect aggressiveness.

2.2.3. The Exposure to Violence Scale (Orue & Calvete, 2008)

The Exposure to Violence Scale has 21 items about the exposure to violence (physical, verbal, and threats) in four contexts: at school, in the neighborhood, at home, and on television. The items include direct exposure, which refers to victimization and indirect exposure, which refers to situations in which the adolescent witnesses violence. Each item is responded on a 5-point Likert-type scale ranging from 0 (*never*) to 4 (*every day*). The Cronbach alpha coefficient was .88.

2.2.4. Peer rejection and acceptance

Following the procedure of most of the previous studies (Dodge et al., 2003; Miller-Johnson, Coie, Maumary-Gremaud, Bierman and Conduct Problems Prevention Research Group, 2002; Prinstein & La Greca, 2004), a peer nomination instrument was used to assess acceptance and rejection by peers. The adolescents indicated the names of the classmates they “liked the most” and those they “liked the least.” The nominations were standardized within each class to obtain the scores of rejection and acceptance for each student. Previous studies indicate that peer nominations and the scores thereby obtained are a valid indication of acceptance and rejection, with good indexes of test–retest validity (Coie & Dodge, 1983). In addition to the sociometric approach, we also assessed perceived social support by means of the *Multidimensional Scale of Perceived Social Support* (MSPSS; Zimet, Dahlem, Zimet, & Farley, 1988). This scale assesses support perceived to be available from family, friends, and significant others. Each item is rated using a scale from 1 (*Very Strongly Disagree*) to 7 (*Very Strongly Agree*). This scale has shown good reliability and validity, with Cronbach’s alpha ranging from .84 to .92 in various American samples (Zimet, Powell, Farley, Werkman, & Berkoff, 1990), and a Cronbach’s alpha of .89 in a Spanish sample (Landeta & Calvete, 2002). In this study, we only used the subscale of Friends’ Support which had a Cronbach alpha coefficient of .81.

2.3. Procedure

The headmasters of several schools were contacted and asked for their schools to participate in the study. After granting permission, they decided to collect passive consent from the students’ parents. Thus, parents were informed and given the option of refusing to allow their son/daughter’s participation. The adolescents filled in the questionnaires in their classrooms. Participants were informed that their responses were confidential and were encouraged to ask questions if they had any trouble answering the items. Participation was voluntary, and a few adolescents ($n = 6$) decided not to participate in the study. The questionnaires took around 50 min to complete.

3. Results

3.1. Factor analysis of the Cyberbullying Questionnaire

The Kaise–Meyer–Olkin index was .96, indicating that the correlation matrix was suitable for factor analysis. The parameters for confirmatory factor analysis were estimated using the polychoric and the asymptotic covariance matrixes of the CBQ items. We tested a one-factor model via Weight Least Squared estimation with LISREL 8.8 (Jöreskog & Sörbom, 2006). Following the recommendations from a number of authors (e.g., Hu & Bentler, 1999), goodness of fit was assessed by the comparative fit index (CFI), the nonnormed fit index (NNFI), and the root mean square error of approximation (RMSEA). Generally, CFI and NNFI values of .95 or above and RMSEA values less than .06 reflect good fit. In all the models, we used the effects-coding method proposed by Little,

Slegers, and Card (2006) to identify and set the scale of the latent variables. This method consists of constraining the set of indicator intercepts to sum to zero for each construct and the set of loadings for a given construct to average 1.0, which is the same as having them sum to the number of unique indicators. According to Little et al., this method is suitable to confirm the factor structure of a construct from particular items. The fit indexes were excellent for the model, $\chi^2(104, n = 1431) = 140$, RMSEA = .016 (0.0079; 0.022), NNFI = 1, CFI = 1. All the factor loadings ranged between .90 and .99. Alpha coefficient was .96 and the mean correlation between items was .64, revealing that, in general, the use of the diverse CB modalities was highly consistent.

3.2. Frequency of CB behaviors

Of the sample, 44.1% of the adolescents responded affirmatively to at least one of the items of the CBQ. Table 2 presents the prevalence rates of each type of CB. The most frequent behaviors were intentionally excluding a classmate from an online group (20.2%), hanging jokes, rumors, gossip, or embarrassing comments about a classmate on the Internet (20.1%), sending the link of such comments to others (16.8%), and hacking in order to send messages by e-mail that could cause trouble for the victim (18.1%). The two forms of CB known as happy slapping (filming someone while they are forced to do something humiliating or filming someone while they are being attacked) were indicated by 10.4 and 10.5%, respectively, of the adolescents.

The content of the responses to the open questions were analyzed. Examples of Items 3 and 4 were “changing clothes in the locker room”, “bathing nude in the river”, “manipulated photo-

graphs, adding, for example, a moustache”, or “silly dancing”. Examples of Items 8 and 9 were “cutting off the leg of a chair so they will fall down when they sit and then recording them” and “making someone sing something silly and sending the video”. Most of the contents of Items 10 and 11 refer to kicking classmates, or, in some cases, a vagabond. Lastly, in some cases, the adolescents indicated that they recorded aggressive scenes but as a joke, like a stage scene. In these cases, it was not considered CB and these responses were not included in the prevalence rate or in the other statistical analyses.

3.3. Variables associated with CB behavior

Gender differences were found in the use of CB. With regard to the prevalence rate, 40.3% of the girls and 47.8% of the boys responded affirmatively to at least one of the CBQ items, and the difference was statistically significant, $\chi^2(1, n = 1403) = 7.95, p < .01$. When assessing the differences by item, we found that they occurred in the behaviors of recording humiliating images of classmates, $\chi^2(2, n = 1380) = 8.45, p < .05$, recording physical aggressions, $\chi^2(2, n = 1380) = 10.41, p < .01$, sending images of physical aggression, $\chi^2(2, n = 1380) = 8.72, p < .05$, and sending images of a sexual nature, $\chi^2(2, n = 1380) = 8.27, p < .05$. The differences were observed especially for the response category “often”. So, 1.8% of the boys versus 0.1% of the girls often recorded physical aggressions, 2% of the boys versus 0.4% of the girls recorded humiliating images of a classmate, 2.14% of the boys versus 0.4% girls sent images of the recorded physical aggressions, and 1.5% of the boys versus 0.1% of the girls sent images of classmates of a sexual nature.

The total frequency of CB was calculated by adding the responses to the 16 CBQ items and an analysis of variance was carried out to assess the differences as a function of gender and school level. Table 3 shows the means and standard deviations obtained. For school grade, statistically significant differences were observed, $F(3, 1390) = 25, p < .001$. Subsequent comparisons (Tukey's method) indicated that the frequency of CB behaviors was significantly higher in second and third grade of secondary education than in first and fourth grade, $p < .05$. There were no significant differences between second and third grade. No significant gender differences were found in CB frequency, $M = 2.15$ and $SD = 5$ in girls versus $M = 2.53$ and $SD = 4.99$ in boys, $F(1, 1390) = 2.69, ns$. However, boys scored higher than girls in all the other aggressive behaviors (proactive, reactive, direct and indirect).

To assess the association between CB behavior and other indicators of aggressive behavior, we carried out multiple regression analysis including the scores in proactive aggressive behavior, reactive aggressive behavior, direct aggressive behavior, indirect/relational aggressive behavior, and justification of violence as predictor variables. This model explained 13% of the variance ($R^2 = .13, p < .001$). Table 4 shows the results. The total CB score was only significantly associated with proactive aggressiveness and beliefs that justify violence.

Lastly, another regression model was estimated, including as predictor variables the contextual variables rejection by others, acceptance by others, perceived social support, and exposure to

Table 2
Prevalence rates per items.

	Sometimes	Often	Total
1. Sending threatening or insulting messages by e-mail	14.2	1.6	15.8
2. Sending threatening or insulting messages by cell phone	14.2	1.5	15.7
3. Hanging humiliating images of classmates on the Internet	9.0	1.0	10
4. Sending links of humiliating images to other people for them to see	8.2	0.9	9.1
5. Writing embarrassing jokes, rumors, gossip, or comments about a classmate on the Internet	18.3	1.8	20.1
6. Sending links with rumors, gossip, etc. about a classmate to other people so they can read them	15.5	1.3	16.8
7. Hacking to send messages by e-mail that could make trouble for the other person	15.5	2.6	18.1
8. Recording a video or taking pictures by cell phone while a group laughs and forces another person to do something humiliating or ridiculous	9.4	1.0	10.4
9. Sending these images to other people	9.6	1.5	11.1
10. Recording a video or taking pictures by cell phone while someone hits or hurts another person	9.6	0.9	10.5
11. Sending these recorded images to other people	10.0	1.3	11.3
12. Broadcasting online other people's secrets, compromising information or images	13.3	1.3	14.6
13. Deliberately excluding someone from an online group	18.1	2.1	20.2
14. Sending messages massively that include threats or are very intimidating	8.3	0.9	9.2
15. Recording a video or taking cell phone pictures of classmates performing some kind of behavior of a sexual nature	8.4	0.7	9.1
16. Sending these images to other people	7.7	0.9	8.6

Table 3
Frequency of CB as a function of academic grade and gender.

	First grade of ESO <i>n</i> = 389	Second of ESO <i>n</i> = 353	Third grade of ESO <i>n</i> = 369	Fourth grade of ESO <i>n</i> = 300	Total
Female	0.76 (1.81)	4.17 (6.98)	2.40 (5.08)	1.19 (1.75)	2.15 (4.73)
Male	1.27 (2.69)	3.41 (5.63)	3.46 (6.28)	2.05 (4.38)	2.53 (5.00)
Mean (SD)	1.02 (2.32)	3.82 (6.39)	2.93 (5.73)	1.56 (3.19)	2.33 (4.86)

Note: ESO, Educación Secundaria Obligatoria (compulsory secondary education).

Table 4

Indicators of aggressive behavior associated with CB.

	<i>B</i>	<i>SE</i>	β	<i>t</i>
Proactive aggressiveness	.52	.06	.34	8.29**
Reactive aggressiveness	-.05	.06	-.04	-0.90
Direct aggressiveness	-.06	.06	-.03	-0.93
Relational aggressiveness	-.00	.07	.00	-0.06
Justification of violence	.07	.03	.08	2.20*

* $p < .05$.** $p < .001$.**Table 5**

Contextual variables associated with CB.

	<i>B</i>	<i>SE</i>	β	<i>t</i>
Rejection by others	.003	.17	.00	0.02
Acceptance by others	.031	.17	.01	0.19
Perceived social support	-.122	.04	-.10	-3.10**
Exposure to violence	.057	.01	.13	4.21**

* $p < .05$.** $p < .001$.

violence. This model only explained 2.5% of the variance of CB ($R^2 = .025$, $p < .001$). Table 5 shows the results. CB was positively associated with exposure to violence and negatively with perceived social support.

4. Discussion

The first goal of the study consisted of developing a questionnaire that included a broad array of CB behaviors. The CBQ includes 16 of such behaviors and has shown excellent psychometric properties insofar as concerns the factor structure and internal consistency.

The most frequent CB modalities were writing embarrassing rumors or comments about classmates on the Internet, sending this link to another person, deliberately excluding someone from an online group, and hacking a classmate in order to send trouble-making e-mail messages to others in the victim's name. The high rate of this last modality is surprising. However, it coincides with the results obtained by Vandebosch and Van Cleemput (2008), who used focus groups and found that one of the most frequently mentioned topics was hacking. Many students admitted that they (or someone they knew) had been a victim of hacking. For example, someone had broken into their MSN account and had changed the password and sent insulting or weird messages to their contacts.

The total prevalence of CB when the *sometimes* and *often* response categories were included was 44.1, which approaches that obtained by Aricak et al. (2008), who found a prevalence rate of 35.7%. However, the prevalence we obtained is much higher than that reported in most previous studies (e.g., Li, 2007a, 2007b; Ortega et al., 2008; Smith et al., 2008). One of the reasons for the high rate obtained in this study may be the instrument employed, which assesses a broad spectrum of CB acts, so there is a higher likelihood that the adolescents will respond affirmatively to at least one type of CB. In accordance with this interpretation, the study of Aricak et al. also used an instrument with many CB items. In fact, in the present study, the prevalence rates of each of type CB ranged between 8.7 (sending images of a classmate of a sexual nature) and 20.2 (deliberately excluding a classmate from an online group). These values are consistent with the prevalence rates obtained in the studies that employed one or a few CB items (e.g., Dehue et al., 2008; Hinduja & Patchin, 2008; Li, 2006; Li, 2007a, 2007b).

Independently of the type of instrument, the data may also express an increase in the problem. When carrying out the study, the headmasters of various centers from which the data were collected

confirmed that they had recently had severe cases of CB. Moreover, the adolescents were not surprised at the questions, but instead they seemed to be familiar with CB. In fact, some adolescents who said they had not participated in any CB acts wrote in the comments section that they had seen other classmates perform such acts.

It is important to note that the prevalence rate of each type of CB behavior is lower when attending only to the response choice *often*, which better matches the conception of CB defined as an aggressive behavior that is *repeated over time*. Nevertheless, we think that carrying out CB acts *sometimes* is also sufficiently severe to be taken into account in the prevalence rates.

The second goal of the study consisted of identifying the cyberbully profile. Coinciding with various previous studies, the boys displayed more frequent use of CB than the girls (e.g., Dehue et al., 2008). Furthermore, we studied differences as a function of the items, observing that such differences were particularly present in the items related to happy slapping. This modality of CB, in contrast to the others, also involves the use of violence in a real setting. This is consistent with previous findings that have found that boys are physically more aggressive than girls, but they use some forms of internet bullying (telling lies about students through e-mail) in the same way (Williams & Guerra, 2007). The results also suggest that the peak frequency of CB is between the second and third year of Secondary Education (13–15 years). Williams and Guerra (2007) also found the highest prevalence in 8th grade (13-year-olds) comparing both with 5th grade (10-year-olds) and 11th grade (16-year-olds).

With regard to the relation of CB with the other indicators of violence, CB was significantly associated with the justification of violence, as reported in the study of Williams and Guerra (2007), and in accordance with the extensive literature that links beliefs in the justification of violence to the use of aggression (Espelage & Swearer, 2003; Huesmann, 1988).

Moreover, as expected, CB was related to the use of proactive aggression, which is carried out coldly and in order to achieve some goal. In this sense, this result is coherent with the studies that conceptualize traditional bullying as essentially proactive aggression (Roland & Idsøe, 2001; Unnever, 2005). However, in contrast to our expectations, CB was not associated with relational aggression. This shares contents with some types of CB, for example, excluding someone from the group or spreading rumors. An explanation of this lack of association may lie in the conditions under which many CB acts are carried out (anonymity, impersonal environment and lack of consequences for the aggressor), which may cause adolescents who would otherwise never dare to perform forms of violence in the school setting to do so through electronic settings (Hinduja & Patchin, 2008). A second explanation of the lack of association with the scores in direct and relational aggression lies in the fact that these scores were based on sociometric measures, whereas CB was assessed by means of self-reports.

In support of this explanation, it is important to note that CB was not associated with the sociometric indicators of rejection and acceptance by others, despite the fact that previous studies of bullying and other forms of aggression have found significant relationships (Card & Little, 2006; Dodge et al., 2006; Laird et al., 2001). More specifically, this result does not support the hypothesis developed by Piazza and Bering (2009), who proposed that relational cyberbullying would be initiated by high-status adolescents. In contrast, perceived peer social support was significantly associated with CB. Adolescents who performed CB were characterized by having a lower perception of social support from their friends. Williams and Guerra (2007) also found that less peer support predicted all the types of bullying including internet bullying. Future research should explore this aspect in more detail because it suggests that isolation and poor social relationships can act as risk factors for CB.

Lastly, CB was significantly related to higher environmental exposure to violence. This result thus joins the extensive bibliography about the role of exposure to violence as a risk factor for learning and developing aggressive behaviors (Baldry, 2003; Flannery et al., 2004; Gorman-Smith et al., 2004).

The main shortcoming of our research is that we relied only on self-report measures to assess CB behavior. Some adolescents may over- or under-report their involvement in CB. Future studies should include peer, parent or teacher reports in order to control for external validity, and take into account different perspectives, such as the reactions of parents and teachers when they discover these behaviors. Moreover, group effects on CB behavior should be addressed. For instance, the risk of CB could be particularly high in some groups of adolescents who consider this behavior appropriate. Namely, qualitative research methods (e.g., focus groups) would provide a useful approach to aspects such as the approval of CB behavior by group members and the dynamics among members that contribute to CB. Finally, in the present study, only the aggressors' profile was studied. Future studies should address the overlapping between CB perpetration and victimization, and the consequences of CB behavior for aggressors and victims.

To conclude, the results of this study warn about increasing online violence among adolescents and indicate that CB is becoming a highly concerning problem in the school setting and for society in general. This is one of the few studies that has assessed the factors associated with performance of CB and it shows that some of the predictors of CB are common to other types of aggressive behavior and should be taken into account with a view to preventing this problem. Specifically, intervention should deal with the normative beliefs about the use of violence, for example, through activities aimed at achieving empathy, and should strengthen interpersonal peer relationships in order to promote a system of positive social support (Olweus & Limber, 1999). Cooperative learning in small groups or common positive activities, such as organizing a trip or a dance where everybody takes part, have been proposed as measures to establish positive peer relationships and prevent bullying (Olweus, 1993; Stevens, De Bourdeaudhuij, & Van Oost, 2001).

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Appendix A

The following sentences refer to the use of the Internet and cell phones. Mark with an "X" the corresponding cells if you have ever carried out any of these actions, using this scale: 0 = never, 1 = sometimes, and 2 = often.

	0	1	2
1. Sending threatening or insulting messages by e-mail			
2. Sending threatening or insulting messages by cell phone			
3. Hanging humiliating images of a classmate on the Internet			
If so, describe the type of images			
4. Sending links of humiliating images to other people for them to see			
If so, describe:			
5. Writing embarrassing jokes, rumors, gossip, or comments about a classmate on the Internet			
6. Sending links with rumors, gossip, etc., of a classmate			

Appendix A (continued)

	0	1	2
or an acquaintance to other people for them to read			
7. Getting someone's password (nicks, cues, etc.) and sending e-mail messages to others in this person's name, which could make this person lose face or cause trouble with his or her acquaintances			
8. Recording a video or taking pictures by cell phone while a group laughs and forces another person to do something humiliating or ridiculous			
If so, describe:			
9. Sending these images to other people			
10. Recording a video or taking pictures by cell phone while someone hits or hurts another person			
If so, describe:			
11. Sending these recorded images to other people			
12. Broadcasting online other people's secrets, compromising information or images			
13. Deliberately excluding someone from an online group (chats, lists of friends, thematic forums, etc.)			
14. Sending messages repeatedly that include threats or that are very intimidating			
15. Recording a video or taking cell phone pictures of some classmate while he or she is carrying out some kind of behavior of a sexual nature			
16. Sending these images to other people			

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